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Optimizing the FINs app using LLMs

The Food Information Networks (FINs) project is designed to enhance the grocery shopping experience. With the rising issues of food deserts in US cities and a lack of access to information to make healthy food choices, people lose motivation to pursue a healthy diet. By leveraging Large Language Models (LLMs), FINs analyzes a user's shopping list alongside stored information about their specific healthy goals, dietary preferences, and a live database of grocery store inventory. Through this, the app will be able to provide users with healthier and financially appropriate alternatives for their grocery shopping.

Technical Contributions and Research Role

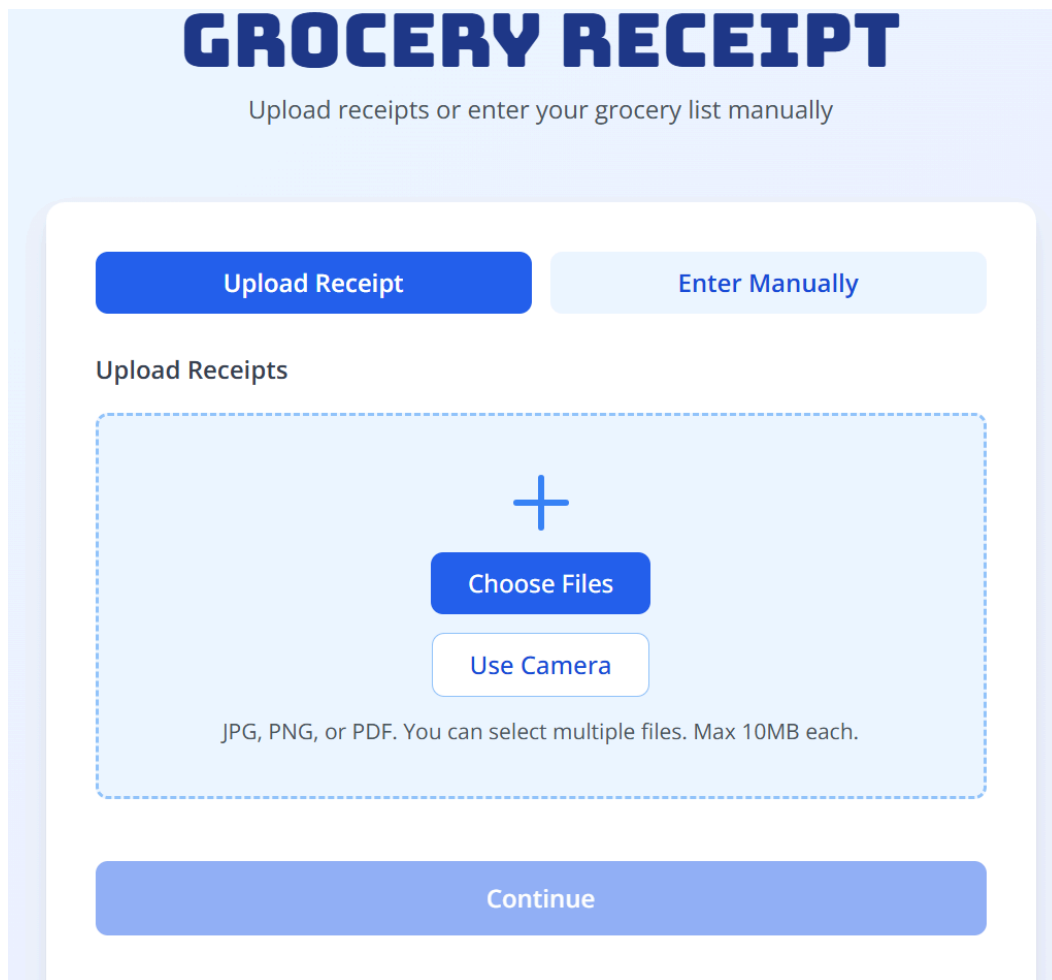
After last semester's work in utilizing the OpenAI API to align AI's logic with the project goals for generating appropriate substitutions and testing the system's ability to process images and analyze user data and input, I focused on the development of a functional web application that aligns with these goals. This included developing a front-end where users can upload receipts, extracting information using the OpenAI API, integrating different user interfaces, and exploring and applying Instacart APIs to shop for substitutions.

Project and Testing Breakdown

Weeks 1-2: We began by learning about various testing techniques and how to use OpenAI for our project. We set up GitHub environments, and learnt more about how the optimization works and the purpose of building a user interface. I started thinking about the best ways to create the substitution lists, analyze user data, and present it to the user.

Week 2-4: Throughout these weeks, we met with Annalisa and understood the goals of the project better. I started by implementing a React web application and building a basic “upload receipt” page. I learnt the importance of thinking about what the user needs and made several changes to the design after weekly meetings with Annalisa and feedback during Wednesday meetings. For example, I explored more user-friendly options for uploading and added one where they can upload pictures through their camera or their device. I also tested different colors and tools for the user to navigate pages.

Week 3-5: I continued building the pages and moved on from an upload page to extract information and selection. First, I used a placeholder for the extract receipt information page, as I focused on giving the users an option to remove and select from the products extracted, while displaying the price and any changes. The main component was the ReceiptUpload.tsx, as shown in figure one, which lets the user either upload receipt files or manually type grocery items, gets back a receiptId, and redirects to the receipt summary page.



The image shows a web interface for uploading grocery receipts. At the top, the title "GROCERY RECEIPT" is displayed in large, bold, dark blue letters. Below the title, a subtitle reads "Upload receipts or enter your grocery list manually". The main content area is a light blue rounded rectangle containing two buttons: "Upload Receipt" (dark blue) and "Enter Manually" (light blue). Below these buttons, the section "Upload Receipts" is titled. It features a large dashed blue box with a blue plus sign in the center. Inside this box are two buttons: "Choose Files" (dark blue) and "Use Camera" (light blue). Below the dashed box, a note states "JPG, PNG, or PDF. You can select multiple files. Max 10MB each." At the bottom of the main content area is a wide, light blue button labeled "Continue".

Figure 1. Grocery Receipt upload page

Week 5-6: At this point, I started testing the OpenAI API to extract information from the receipts and to create the receipt summary page, as explained above. The main component was `ReceiptExtraction.tsx`, which is the review screen shown in Figure 2. It loads the `receiptId` from the URL or session storage that is saving data between pages, fetches the extracted receipt data from `/api/extractReceipt`, and displays the products in a table.

Select which products should be included in your score. Removing deletes them from this list.

Include	Item	Quantity	Price	Action
☒	CHECKLANE CAND	1	\$1.39	Remove
☒	MILK	1	\$2.63	Remove
☒	EGGS	1	\$0.00	Remove
☒	STOK COLD BREW	1	\$6.39	Remove
☒	SAUSAGE ROLL	1	\$0.00	Remove
☒	THOMAS BAGELS	1	\$3.99	Remove

All items total:
 \$14.40

Included in score:
 \$14.40

Upload Another Receipt

Find Nearby Stores

Figure 2. Receipt Summary Page

Week 7-8: After the mid-semester check-in and with Professor Metoyer's and Annalissa's revision, I started exploring the ways we can utilize the Instacart API to improve user experience and the process of providing alternatives clearly. I researched how the API handled store and location data, experimented with integrating it into the interface, and considered how it would make the application more user-friendly. At that time, we faced some technical difficulties with GitHub, which prevented me from accessing and pushing code, so I continued focusing on learning more about the API and its applications. At the end of these two weeks, the goals of implementing the API were still under research, because of the limitations of the API.

Week 8-9: I implemented the first part of the InstaCart API, which finds nearby stores using a ZIP code. It is based on `InstacartPageContent.tsx`, which reads the selected receipt items from session storage, lets the user enter a ZIP code, and calls the Instacart retailer API to find nearby stores. As shown in Figure 3, it first shows the receipt items and gives the option to go back to the receipt. During those weeks, we had a discussion about how users can navigate different pages, and I started working on ways to make navigation easier. I added buttons like the “Back to Receipt” and the “Review shopping list” so that the number of items and the names of products are clearer. Then, with the API, the user can open the final shopping list in Instacart, where products are copied, the store nearby is selected, and the items are displayed, as shown in Figure 4. At that point, we were also exploring how to integrate the alternatives and add them to the web app.

We can show nearby Instacart retailers for your ZIP code, let you pick a preferred store, and preview a richer shopping list before sending you to Instacart.

Back to receipt

Receipt items

CHECKLANE CAND

MILK

EGGS

STOK COLD BREW

SAUSAGE ROLL

THOMAS BAGELS

1 Enter ZIP code

Find nearby Instacart retailers in your area before opening the shopping list.

ZIP code

46556

Find nearby stores

2 Nearby retailers

Instacart will confirm the final store on its side, but these are the retailers available near your ZIP code.

Jewel-Osco
jewel-osco

ALDI
aldi

Kroger

Strack & Van Til

Figure 3. Receipt Items and Nearby Stores

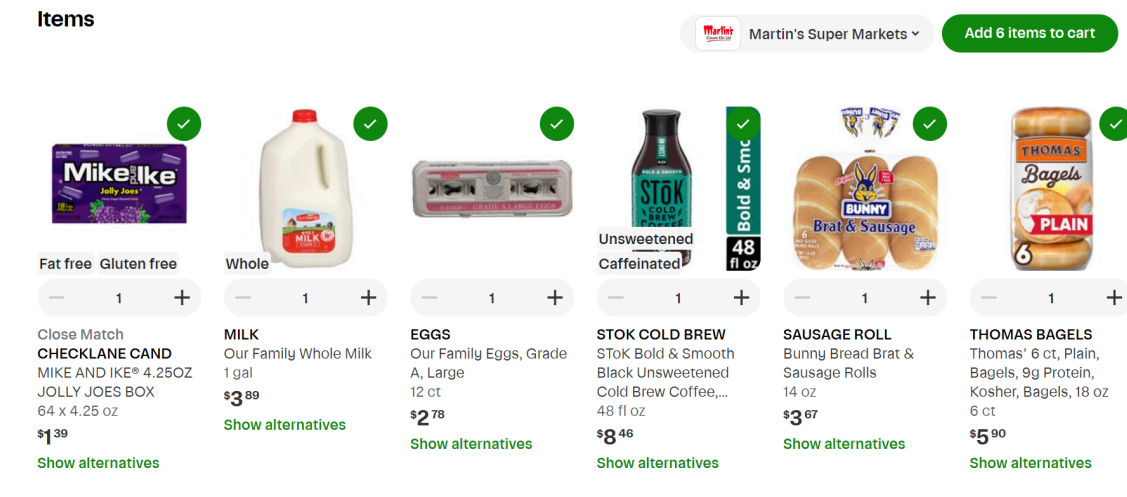


Figure 4. InstaCart Page

Week 9-10: I utilized more of InstaCart's API to be able to classify the categories of the products and show the InstaCart Search and the best aisle to be found. Ideally, I was researching ways to display more information about products, but due to limited InstaCart API options, the user is only able to view full information and buy on InstaCart website.

Week 11-end of semester: At this point in the semester, I focused on refining the project and organize the work completed throughout the term. I reviewed feedback and considered possible improvements in implementing the InstaCart API. I revisited some areas to refine the user experience and worked on the demo and thought about exploring how to use multiple Instacart features together to make the shopping lists even smarter and the next steps in merging different team programs and ensuring a smooth user experience during testing

Reflections and Skills Gained

The area of Human-Computer Interaction is of high importance and interest to me. This semester, I got to learn how to think about users and how the intersection between user data, preferences, and AI all correlate together. Combining the aspect of dietary choices with how AI

could potentially enhance better (from both a healthy and financial perspective) substitutions to match those choices opened my eyes to the precision needed in results and the numerous experiments it requires to achieve them. From building the React web application, I learnt about user experience and implementing APIs. The feedback and the challenges taught me how to build a better interface and to integrate as many tools to make it meet the needs of the user. I am also very glad that I got to work under mentorship, receive feedback, and grow with what I learn, and I am passionate about continuing this process. I learned to collaborate with others in testing, presentation, and most importantly, the process of analyzing results and decision making, which is an aspect I am passionate about in Engineering projects.

Areas of Improvement

While all the different components of the project are significant, I wish we could have integrated everyone's work and analyzed how the web application becomes a full process, including providing alternatives. Moreover, if there were more time, we could have explored other APIs and ways to make substitutions clearer to the user. These would have been the last steps in turning the project into a complete interactive web app.